Snapdragon Sense: AI-Powered Laptop Troubleshooting

Prototype AI assistant designed to diagnose and resolve laptop issues locally, ensuring privacy and efficiency on Snapdragon-powered HP PCs.



User Problem

Diagnosing laptop issues is **complex** and **time-consuming**

Limited access to expert support without exposing private data

Dependence on cloud services raises **privacy** and **latency** concerns



FRUSTRATED USER

COMPLEX ERROR MESSAGES

PRIVACY & LATENCY RISKS

Proposed Solution

An AI-powered troubleshooting assistant that delivers accurate, **private,** and reliable laptop support—built for the future.



AI-DRIVEN DIAGNOSIS

Analyzes user-provided or simulated device metrics locally.

INTERACTIVE UI

Streamlit-based interface for simple and guided user interaction.

SAFE FALLBACK

Rules-based backup ensures dependable guidance when AI is uncertain.

BUILT FOR PRIVACY & EFFICIENCY

Designed for future Snapdragon-powered HP laptops with privacy and efficiency at the core.

User Journey & Demo

1

Launch

User opens the Streamlit interface on their laptop

2

Share

User enters symptoms or collects system metrics

3

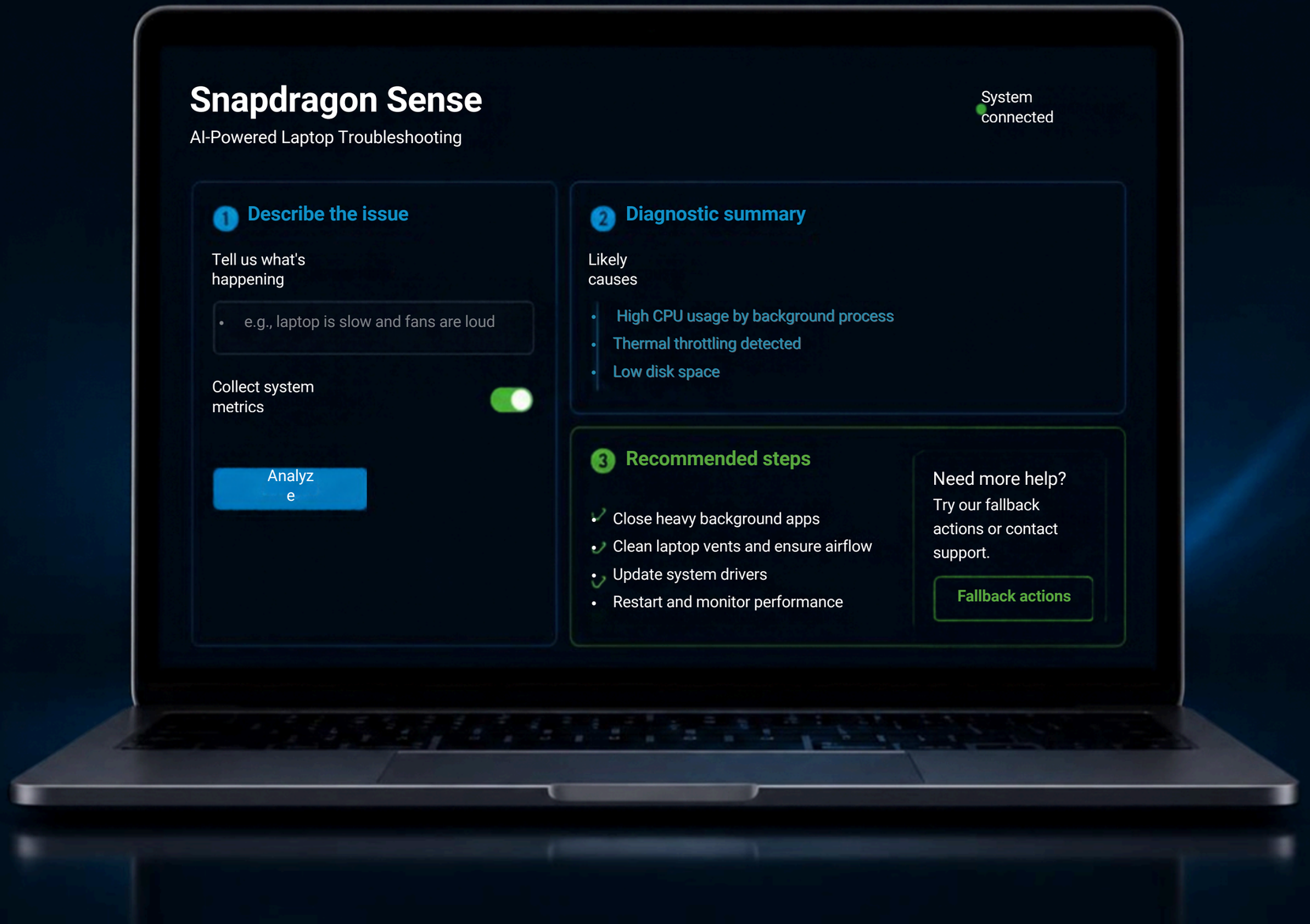
Diagnose

AI model analyzes data and returns likely root causes

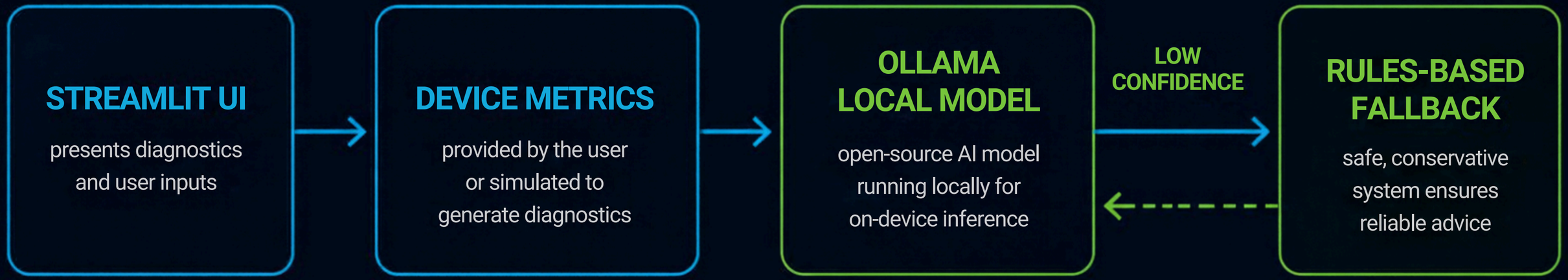
4

Resolve

User follows guided steps or fallback advice to fix



AI & Technical Architecture



INTERACTIVE FRONTEND
central place for diagnostics and user interaction

INPUTS
real device metrics or simulated data power diagnostic generation

AI ANALYSIS
local model analyzes metrics and generates clear, actionable guidance

SAFEGUARD
fallback logic provides reliable recommendations when confidence is low

Snapdragon Fit



ON-DEVICE AI

Built for on-device inference, reducing reliance on the cloud.

PRIVACY FIRST

Data stays on your device, protecting user privacy.

LOW POWER

Optimized for efficiency on Snapdragon-powered HP laptops.

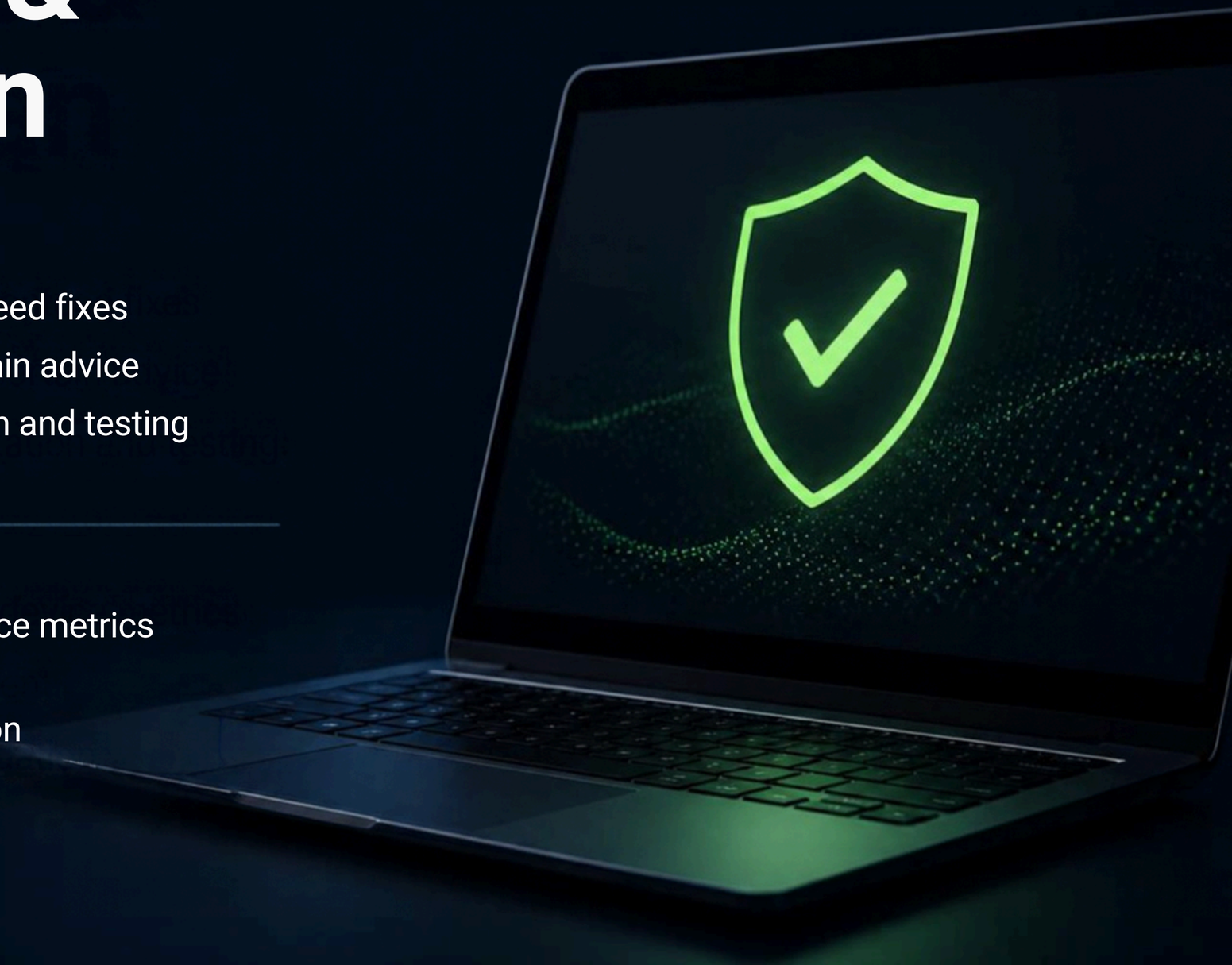
FUTURE VALIDATION

Performance will be validated and fine-tuned on Snapdragon hardware.

Safety, Limits & Validation Plan

Ai provides diagnostic suggestions, not guaranteed fixes
Rules-based fallback prevents unsafe or uncertain advice
Prototype status emphasizes future optimization and testing

Simulated environment testing with diverse device metrics
User **feedback** cycles for iterative improvement
Planned **performance** and compatibility validation
on Snapdragon HP devices



Impact & Next Steps

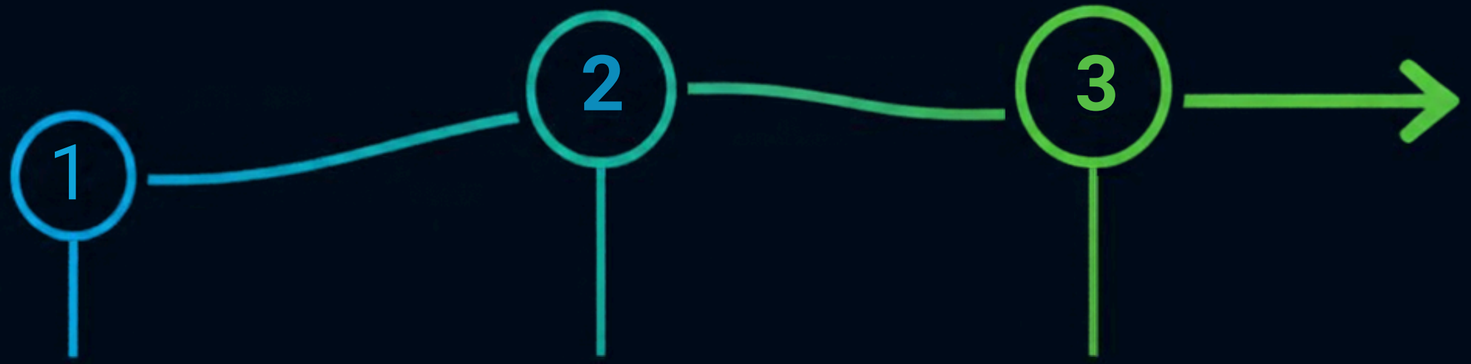
IMPACT

Empowers users with private, efficient troubleshooting

Reduces support costs and improves device uptime

Advances AI integration on Snapdragon platforms

NEXT STEPS



1. Optimize AI models for Snapdragon on-device inference
2. Develop richer diagnostics and expand device coverage
3. Collaborate with HP and Snapdragon teams for validation and deployment